

PCA AND ITS APPLICATION IN ESTIMATING RISK PREMIA

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1. RISK PREMIA

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- Characteristic-based prediction model: bottom-up

Recall the basic model setup in our class (discussing ML in return prediction):

$$r_t = \mu + \epsilon_t, \quad \mu = X_t g$$

$$\begin{bmatrix} r_{11} & \cdots & r_{1T} \\ r_{21} & \cdots & r_{2T} \\ \vdots & \ddots & \vdots \\ r_{N1} & \cdots & r_{NT} \end{bmatrix}_{N \times T} = \begin{bmatrix} x_{11} & \cdots & x_{1K} \\ x_{21} & \cdots & x_{2K} \\ \vdots & \ddots & \vdots \\ x_{N1} & \cdots & x_{NK} \end{bmatrix}_{N \times K} \begin{bmatrix} g_{11} & g_{12} & \cdots & g_{1T} \\ g_{21} & g_{22} & \cdots & g_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ g_{K1} & g_{K2} & \cdots & g_{KT} \end{bmatrix}_{K \times T} + \begin{bmatrix} \epsilon_{11} & \cdots & \epsilon_{1T} \\ \epsilon_{21} & \cdots & \epsilon_{2T} \\ \vdots & \ddots & \vdots \\ \epsilon_{N1} & \cdots & \epsilon_{NT} \end{bmatrix}_{N \times T}$$

- Factor pricing model: top-down

$$r_t = \beta f_t + u_t$$

$$\begin{bmatrix} r_{11} & \cdots & r_{1T} \\ r_{21} & \cdots & r_{2T} \\ \vdots & \ddots & \vdots \\ r_{N1} & \cdots & r_{NT} \end{bmatrix}_{N \times T} = \begin{bmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1p} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{N1} & \beta_{N2} & \cdots & \beta_{Np} \end{bmatrix}_{N \times p} \begin{bmatrix} f_{11} & \cdots & f_{1T} \\ f_{21} & \cdots & f_{2T} \\ \vdots & \ddots & \vdots \\ f_{p1} & \cdots & f_{pT} \end{bmatrix}_{p \times T} + \begin{bmatrix} u_{11} & \cdots & u_{1T} \\ u_{21} & \cdots & u_{2T} \\ \vdots & \ddots & \vdots \\ u_{N1} & \cdots & u_{NT} \end{bmatrix}_{N \times T}$$

1. RISK PREMIA

What is "risk premia"?

- The risk premium of factors $f = (f_1, \dots, f_K)^T$ is given by (Ross, 1976)

$$\mathbb{E}[r_i] = \beta_i^T \gamma$$

where $\gamma = (\gamma_1, \dots, \gamma_K)^T$ denotes the vector of factor risk premia.

- Risk premium γ_k is interpreted as expected compensation for bearing one additional unit of exposure to factor f_k .
- Always prefer γ to be expressed in return terms.

Then, how can we represent "risk premium" in the factor pricing model?

- Tempted to think that

$$\gamma = \mathbb{E}[f_t]$$

1. RISK PREMIA

However,

- Some factors $\{f_i\}$ are measured in returns, e.g. market portfolio return.
- Some $\{f_j\}$ are not, e.g. climate risk or inflation.
 - In these cases, we cannot directly interpret $\mathbb{E}(f_j)$ as risk premium.

Instead, we modify the model by decomposing the factor into two parts:

- γ : captures the risk premia for factors f_t , i.e. the mean of shocks expressed in return terms.
- v_t : captures the variance of shocks (factors) that affect asset returns, i.e. demeaned f_t .

Thus, to estimate the risk premia, the factor pricing model is presented by

- when each component of f_t is a tradable factor

$$r_t = \beta f_t + u_t;$$

- when each component of f_t is a non-tradable factor

$$r_t = \beta \gamma + \beta v_t + u_t.$$

1. RISK PREMIA

In the remaining discussion, we focus only on models with non-tradable factors since models with tradable factors are much easier.

$$r_t = \beta\gamma + \beta v_t + u_t$$

The next question is: how to estimate risk premia γ ?

- Most prominently,
 - Fama-MacBeth regressions (Fama & MacBeth, 1973)
 - Mimicking-portfolio projections (Huberman, Kandel & Stambaugh, 1987)

1. RISK PREMIA

For simplicity, suppose there are two non-tradable factors.

The model that we focus on can be presented in the way below:

Way 1 to estimate:

Use T equations to estimate 2 parameters per time.

$$\begin{pmatrix} r_{11} & r_{12} & \dots & r_{1T} \\ r_{21} & r_{22} & \dots & r_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N1} & r_{N2} & \dots & r_{NT} \end{pmatrix}_{N \times T} = \begin{pmatrix} \beta_{11} & \beta_{12} \\ \vdots & \vdots \\ \beta_{N1} & \beta_{N2} \end{pmatrix}_{N \times 2} \begin{pmatrix} \delta_1 & \dots & \delta_1 \\ \delta_2 & \dots & \delta_2 \\ \vdots & \ddots & \vdots \\ \delta_T & \dots & \delta_T \end{pmatrix}_{2 \times T} + \begin{pmatrix} u_{11} & u_{12} & \dots & u_{1T} \\ u_{21} & u_{22} & \dots & u_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ u_{N1} & u_{N2} & \dots & u_{NT} \end{pmatrix}_{N \times T}$$

Way 2 to estimate:

Use N equations to estimate $2N$ parameters per time.

$$\begin{pmatrix} r_{11} & r_{12} & \dots & r_{1T} \\ r_{21} & r_{22} & \dots & r_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N1} & r_{N2} & \dots & r_{NT} \end{pmatrix}_{N \times T} = \begin{pmatrix} \beta_{11} & \beta_{12} \\ \vdots & \vdots \\ \beta_{N1} & \beta_{N2} \end{pmatrix}_{N \times 2} \begin{pmatrix} \delta_1 & \dots & \delta_1 \\ \delta_2 & \dots & \delta_2 \\ \vdots & \ddots & \vdots \\ \delta_T & \dots & \delta_T \end{pmatrix}_{2 \times T} + \begin{pmatrix} u_{11} & u_{12} & \dots & u_{1T} \\ u_{21} & u_{22} & \dots & u_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ u_{N1} & u_{N2} & \dots & u_{NT} \end{pmatrix}_{N \times T}$$

FAMA-MACBETH REGRESSIONS

- Motivation: directly estimate using econometric methods.

Step 1: Time series regressions

- Regress each test asset's excess return onto the factors
- Obtain risk exposures $\hat{\beta}_1, \hat{\beta}_2$

$$\begin{aligned}
 & \text{for } i=1, 2, \dots, N \\
 & \begin{pmatrix} r_{i1} \\ r_{i2} \\ \vdots \\ r_{iT} \end{pmatrix} = \begin{pmatrix} b_i \\ b_i \\ \vdots \\ b_i \end{pmatrix} + \beta_{i1} \begin{pmatrix} v_{11} \\ v_{12} \\ \vdots \\ v_{1T} \end{pmatrix} + \beta_{i2} \begin{pmatrix} v_{21} \\ v_{22} \\ \vdots \\ v_{2T} \end{pmatrix} + \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iT} \end{pmatrix} \\
 & = \begin{pmatrix} 1 & v_{11} & v_{21} \\ 1 & v_{12} & v_{22} \\ \vdots & \vdots & \vdots \\ 1 & v_{1T} & v_{2T} \end{pmatrix} \begin{pmatrix} b_i \\ \beta_{i1} \\ \beta_{i2} \end{pmatrix} + \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iT} \end{pmatrix}
 \end{aligned}$$

Step 2: Cross-sectional regressions

- Regress cross-sectional returns onto $\hat{\beta}_1, \hat{\beta}_2$
- Obtain $\hat{\gamma}_1, \hat{\gamma}_2$

$$\begin{aligned}
 & \text{for } j=1, 2, \dots, T \\
 & \begin{pmatrix} r_{1j} \\ r_{2j} \\ \vdots \\ r_{Nj} \end{pmatrix} = \begin{bmatrix} \hat{\beta}_1 & \hat{\beta}_2 \end{bmatrix}_{N \times 2} \begin{pmatrix} v_{1j} \\ v_{2j} \end{pmatrix}_{2 \times 1} + \begin{pmatrix} u_{1j} \\ u_{2j} \\ \vdots \\ u_{Nj} \end{pmatrix}_{N \times 1} \\
 & \hat{\gamma}_1 = \frac{1}{T} \sum_{j=1}^T \hat{r}_1^{(j)} \\
 & \hat{\gamma}_2 = \frac{1}{T} \sum_{j=1}^T \hat{r}_2^{(j)}
 \end{aligned}$$

MIMICKING-PORTFOLIO PROJECTION

- Motivation:

- Easy to estimate risk premia for tradable factors, since

$$\gamma = \mathbb{E}(f_t)$$

- Whether it is possible to use tradable factors to mimic a non-tradable factor?

- Step 1: Project factor innovation onto asset returns

$$v_t = a + w^T r_t + \eta_t$$

- Step 2: Define mimicking portfolio return

$$r_t^{mp} = w^T r_t$$

$$\gamma = \mathbb{E}(r_t^{mp})$$

2. PROBLEM

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Both Fama-MacBeth regressions (Fama & MacBeth, 1973) and mimicking-portfolio projections (Huberman, Kandel & Stambaugh, 1987) suffer from potential biases induced by

- omitted factors
- measurement error

Setups:

$$r_t = \beta\gamma + \beta v_t + u_t$$
$$\mathbb{E}(v_t) = \mathbb{E}(u_t) = 0, \quad \text{Cov}(u_t, v_t) = 0$$

However, v_t cannot be fully observed.

Let g_t be a set of d observable factors whose risk premia we aim to estimate. Then,

$$g_t = \delta + \eta v_t + z_t$$
$$\mathbb{E}(z_t) = 0, \quad \text{Cov}(z_t, v_t) = 0$$

Thus, the objective of our analysis is the risk premia of g_t :

$$\gamma_g = \eta\gamma$$

3. MOTIVATIONS

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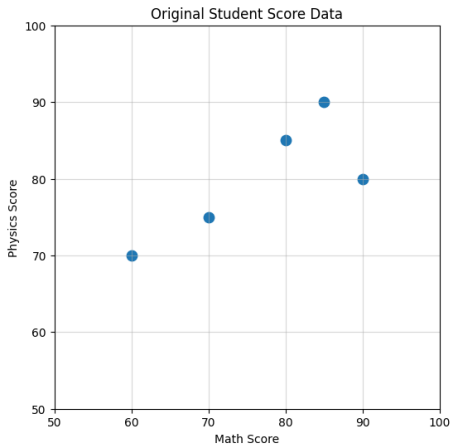
How to find these v_t ?

- Why PCA? (e.g., Connor & Korajczyk, 1986; Bai, 2003)
 - Intuition: Asset returns are driven by a large number of potentially high-dimensional economic forces.
 - But our model assumes that their co-movement can be summarized by a small number of latent factors.
 - In other words, we seek dimensionality reduction.
- What is PCA?
 - A dimensionality reduction method
 - Sequentially identifies multiple orthogonal directions (how many is chosen by us)
 - These directions capture a large share of variation in the data

These papers demonstrate that by applying PCA to a large cross-section of observable asset returns, one can recover an appropriate set of p latent factors that span the underlying factor space.

3. MOTIVATIONS

An Example for PCA



BAI & NG (2003)

- Other literatures: either N or T is fixed; rely on sequential limits; require covariance stationarity and homoskedasticity
- Bai & Ng (2003)

$$\begin{matrix} R & = & \beta & V & + & U \\ (N \times T) & & (N \times p) & (p \times T) & & (N \times T) \end{matrix}$$

- How to estimate $\hat{v}_t$ and $\hat{\beta}$?

Suppose p ($p < \min\{N, T\}$) is given, then the estimates of β^p and V^p are obtained by solving the optimization problem:

$$\begin{aligned} Q(p) &= \min_{\beta, V^p} (NT)^{-1} \sum_{i=1}^N \sum_{t=1}^T (R_{it} - \beta_i^p \cdot V_t^p)^2 \\ &= \min_{\beta, V^p} (NT)^{-1} \|R - \beta V\|^2 \end{aligned}$$

subject to

$$\frac{VV'}{T} = I_p$$

- concentrate out β^p :

If V^p is fixed, by FOC,

$$\hat{\beta}^p = RV'(VV')^{-1} = \frac{R(V^p)'}{T}$$

Substitute back,

$$\max_{V^p} \text{tr}(V^p R^T R (V^p)^T)$$

- Thus,

$$\hat{V} = T^{\frac{1}{2}}(\xi_1 : \xi_2 : \dots : \xi_p)^T$$

$$\hat{\beta} = T^{-1} R \hat{V}^T$$

where $\xi_1, \dots, \xi_{\hat{p}}$ are normalized eigenvectors corresponding to the largest $\hat{p}$ eigenvalues of the matrix

$$n^{-1} T^{-1} R^T R$$

- How to estimate $\hat{p}$?

Much more complex. How to select criteria depends on N and T (Bai & Ng, 2003).

3. MOTIVATIONS

But a new problem is coming.

- PCA factors are only identified up to rotation, therefore individual factors may be economically uninterpretable.
- This problem is tractable by the property of “rotation invariance”.

Using PCA, we can observe $\hat{v}_t = Hv_t$, with H any full-rank $p \times p$ matrix, but observe neither H nor v_t . We rewrite the model:

$$\begin{aligned}r_t &= \beta H^{-1}H\gamma + \beta H^{-1}Hv_t + u_t := \hat{\beta}\hat{\gamma} + \hat{\beta}\hat{v}_t + u_t, \\g_t &= \delta + \eta H^{-1}Hv_t + z_t := \delta + \hat{\eta}\hat{v}_t + z_t.\end{aligned}$$

Since now both $\hat{v}_t$ and g_t are observable, and $\hat{\beta}$ is also known, it is very easy to estimate $\hat{\gamma}$ and $\hat{\eta}$.

Besides, we are surprised to discover that

$$\gamma_g = \eta H^{-1}H\gamma = \hat{\eta}\hat{\gamma},$$

which can be estimated now. The good property on γ_g (only on γ) is called “rotation invariance”.

4. THE THREE-PASS ESTIMATOR

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Notation:

Diagram illustrating the decomposition of a matrix R into a product of matrices Φ , Γ , Z , and V , plus a diagonal matrix U .

The matrices are defined as follows:

- R is an $N \times T$ matrix.
- Φ is an $N \times P$ matrix.
- Γ is a $P \times T$ matrix.
- Z is an $N \times T$ matrix.
- V is an $N \times T$ matrix.
- U is an $N \times T$ matrix.

The decomposition is given by:

$$R = \Phi \Gamma Z^T + V U$$

The diagram also shows the decomposition of the matrix Γ into a product of matrices S and G , plus a diagonal matrix H .

The matrices are defined as follows:

- Γ is a $P \times T$ matrix.
- S is a $P \times T$ matrix.
- G is a $P \times T$ matrix.
- H is a $P \times T$ matrix.

The decomposition is given by:

$$\Gamma = S G + H$$

The diagram also shows the decomposition of the matrix H into a product of matrices S and G , plus a diagonal matrix H .

The matrices are defined as follows:

- H is a $P \times T$ matrix.
- S is a $P \times T$ matrix.
- G is a $P \times T$ matrix.
- H is a $P \times T$ matrix.

The decomposition is given by:

$$H = S G + H$$

- Step 1: PCA

- Conduct the PCA of the matrix $n^{-1}T^{-1}\bar{R}^T\bar{R}$.
- Define the estimator for the factors and their loading as

$$\hat{V} = T^{\frac{1}{2}}(\xi_1 : \xi_2 : \dots : \xi_{\hat{p}})^T$$

$$\hat{\beta} = T^{-1}\bar{R}\hat{V}^T,$$

where $\xi_1, \dots, \xi_{\hat{p}}$ are normalized eigenvectors corresponding to the largest $\hat{p}$ eigenvalues of the matrix $n^{-1}T^{-1}\bar{R}^T\bar{R}$.

- Step 2: Cross-sectional regression

- Run a cross-sectional OLS regression of $\bar{R}$ onto the estimated factor loadings $\hat{\beta}$,

$$\hat{\gamma} = (\hat{\beta}^T\hat{\beta})^{-1}\hat{\beta}^T\bar{R}.$$

- Step 3: Time series regression

- Run a time series regression of g_t onto the extracted factors from step 1

$$\hat{\eta} = \bar{G}\hat{V}^T(\hat{V}\hat{V}^T)^{-1},$$

$$\hat{G} = \hat{\eta}\hat{V}$$

The final goal for us is $\hat{\gamma}_g = \hat{\eta}\hat{\gamma} = \bar{G}\hat{V}^T(\hat{V}\hat{V}^T)^{-1}(\hat{\beta}^T\hat{\beta})^{-1}\hat{\beta}^T\bar{R}$.

5. EMPIRICS

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Data:

- a large set of 647 portfolios
 - US equities, treasury bonds, corporate bonds, and currencies
 - sorted by a large number of characteristics
 - Portfolios rather than individual stocks
- sample covers the period 1976–2010 + monthly frequency
 - $35 \times 12 = 420 < 647 \Rightarrow \hat{p} = 7$
- factors
 - tradable factors: market, size, value, profitability, investment, momentum, betting against beta (Frazzini & Pedersen, 2014) and quality minus junk (Asness, Frazzini & Pedersen, 2013)
 - non-tradable factors: AR(1) innovations in IP growth, VAR(1) innovations in the first three PCs of 279 macrofinance variables from Ludvigson & Ng (2010), the liquidity factor (Pástor & Stambaugh, 2003)...

5. EMPIRICS

TABLE B1
THREE-PASS REGRESSION: EMPIRICAL RESULTS

FACTORS	AVERAGE RETURN		TWO-PASS REGRESSION						MIMICKING PORTFOLIO				THREE-PASS REGRESSION			R^2_{γ}	p (gWeak)
	γ	SE	No Controls		With R_m		With FF3		With R_m		With FF3		γ	SE	R^2_{γ}		
			γ	SE	γ	SE	γ	SE	γ	SE	γ	SE					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	
Market	.51**	.23	.56**	.23	.56**	.23	.51**	.22	.51**	.22	.51**	.22	.51**	.23	99.57	.00	
SMB	.25	.15	.82**	.34	.07	.16	.10	.16	.08**	.04	.25	.16	.20	.16	97.24	.00	
HML	.35**	.17	-.85**	.38	.30*	.18	.35**	.17	-.13**	.06	.35**	.15	.20	.15	83.03	.00	
MOM	.69***	.24	-.20**	.88	.20	.26	.71***	.24	-.05	.05	-.21*	.11	.49**	.23	89.82	.00	
RMW	.38***	.13	.04	.16	-.00	.17	.27**	.13	-.07**	.03	-.09	.06	.22*	.11	71.48	.00	
CMA	.32***	.11	-.59**	.24	.34**	.14	.42***	.12	-.10**	.05	.12	.08	.14	.10	59.03	.00	
BAB	.94***	.22	-1.59*	.85	1.10***	.29	1.21***	.27	-.06	.05	.23**	.11	.57***	.15	47.43	.00	
QMJ	.44***	.14	-.50**	.21	.01	.16	.25*	.14	-.15**	.07	-.29***	.09	.06	.13	84.29	.00	
Liquidity			.226**	.90	3.44***	1.09	.57	.68	.21*	.11	.32**	.14	.37**	.16	12.11	.00	
Intermediary (He)			1.01**	.45	.19	.49	.43	.45	.57**	.25	.78***	.27	.60**	.31	69.05	.00	
Intermediary (Adrian)			1.37***	.30	1.52***	.28	1.58***	.27	.10*	.06	.61***	.15	.72***	.16	51.99	.00	
New York temperature			-319.01	255.73	125.89	152.76	-277.96**	124.08	-2.35	5.42	10.71	10.94	-.69	13.90	.76	.84	
Global temperature			-6.65	4.85	-5.29	4.92	-3.33	2.07	-.01	.09	.11	.17	.05	.21	2.21	.09	
El Niño			56.85***	17.42	19.23*	11.08	-15.34**	7.11	.39	.33	.94	.59	.41	.82	1.58	.43	
Sunspots			-409.37	937.73	1,637.60***	467.40	882.89**	405.40	-19.30	19.49	-4.33	30.42	4.01	35.63	.86	.72	
IP growth			-.36**	.14	-.27***	.07	-.14***	.05	-.00	.00	-.01	.01	-.01*	.00	2.25	.21	
Macro PC 1			84.90***	24.76	87.26***	20.95	39.96***	13.57	1.22	.75	2.49*	1.43	3.26**	1.58	2.34	.29	
Macro PC 2			9.35	15.93	9.28	16.34	23.91**	8.97	-.91	.59	-2.05**	1.03	-.88	1.27	4.05	.09	
Macro PC 3			-5.94	14.30	-6.70	12.11	-31.24***	9.74	-.99	.64	-.61	1.21	-1.25	1.51	6.60	.01	
Consumption growth			.26*	.16	-.03	.11	.07	.05	-.00	.00	-.00	.01	.00	.01	4.07	.07	
Stockholder consumption			6.26***	2.14	2.48**	1.20	1.08*	.58	.05	.04	.03	.06	.17**	.08	2.50	.32	

6. SUMMARY

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- Risk premia: definition and identification
 - Ross, 1976
 - Fama & MacBeth, 1973; Huberman, Kandel & Stambaugh, 1987
- Omitted variable bias in risk premia estimation: PCA as a solution
 - Connor & Korajczyk, 1986; Bai & Ng, 2003
- PCA-based estimation of risk premia
 - Giglio & Xiu, 2021
- Empirical evidence from US stock markets
- Contribution: extensions to course materials
 - Machine learning applications in **factor models**
 - Focus on a topic: how to estimate **risk premia?**